Asia-Pacific Journal of Oncology Nursing 12 (2025) 100746

Contents lists available at ScienceDirect

Asia-Pacific Journal of Oncology Nursing

journal homepage: www.apjon.org

Review

Interventions targeted to improve lung cancer screening uptake among high-risk individuals: A systematic review and meta-analysis

Yuan Yang, Wenqian Zhao, Dorothy N.S. Chan, Shishuang Zhou, Kai Chow Choi, Winnie K.W. So *

The Nethersole School of Nursing, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong, China

ARTICLE INFO

Keywords:
Lung cancer
Cancer screening
High-risk individuals
Systematic review
Meta-analysis

ABSTRACT

Objective: To synthesise evidence on interventions aimed at increasing lung cancer screening uptake among high-risk individuals, and examine the effectiveness of each intervention category.
Methods: Eight databases, namely EMBASE, CINAHL, MEDLINE, Scopus, PubMed, Web of Science, Wan Fang Database, and China National Knowledge Infrastructure, were systematically searched from inception to June 2024. The reference lists of identified studies were manually searched. Two reviewers independently assessed eligibility, extracted data, and evaluated methodological quality using the Cochrane risk of bias 2.0 tool. Meta-analyses and descriptive analyses were conducted. Subgroup analyses were performed for intervention category. All analyses were conducted using RevMan 5.4.
Results: A total of 13 trials from 12 studies were included. Six categories of interventions were identified: online education, patient navigation, shared decision-making, outreach promotion, targeted invitation materials, and combined intervention. The meta-analysis findings revealed that lung cancer screening uptake was enhanced among high-risk individuals receiving various interventions (OR, 1.70; 95% CI, 1.10–2.62; $P = 0.02$) compared with no intervention or usual care. Subgroup analysis indicated that combined intervention (OR, 3.38; 95% CI, 2.08–5.52; $P < 0.001$) and patient navigation (OR, 3.31; 95% CI, 1.70–6.41; $P < 0.001$) could be effective strategies to improve lung cancer screening uptake among high-risk individuals.
Conclusions: Combined intervention and patient navigation are most effective in increasing lung cancer screening uptake among high-risk individuals. However, the optimal design, tailoring, and delivery methods remain unclear due to the limited number of studies included and require further research.

Introduction

Lung cancer is the most common cancer and the leading cause of cancer deaths worldwide, accounting for 12.4% of all cancer cases and 18.7% of all cancer-related deaths.[1] Although lung cancer treatments have improved with health reforms and technological advances, the overall 5-year survival rate remains poor, ranging from 10% to 20% in most regions.[2] However, lung cancer patients can have a favourable prognosis if detected early. A clinical study reported the 5-year survival rate for early-stage lung cancer can reach 61.2%.[3] However, more than half of lung cancer cases are diagnosed at advanced stages, where the 5-year survival rate drops to 8%.[4] Therefore, early detection is critical for improving lung cancer prognosis.

Lung cancer screening (LCS) is a preventive measure designed to detect lung cancer early before symptoms appear, including low-dose computed tomography (LDCT), chest X-ray, and sputum cytology. Among these, LDCT is the most widely recommended approach in counties like the United States and China, as LDCT has higher sensitivity (84.6%–93.0%) than that of chest X-ray (77%–80%).[5–7] In addition, LDCT can significantly reduce lung cancer mortality by 17.2%–20.0% compared with chest X-ray.[5,6] Current guidelines in countries like the USA, Canada, Japan, Korea, China, and the UK recommend annual LDCT for high-risk individuals to maximize effectiveness while minimizing adverse effects like false positives, radiation exposure, and overdiagnosis.[8–10]

Despite robust evidence and guidelines recommending LCS for high-risk individuals, the adoption rate remains low. For example, only nine states in the USA have reported LCS participation rates above 10%, despite widespread recommendations since 2013.[11] A multi-centre population-based study in China found the uptake rate of LCS to be

* *Corresponding author.*
E-mail address: winnieso@cuhk.edu.hk (W.K.W. So).

https://doi.org/10.1016/j.apjon.2025.100746
Received 2 January 2025; Accepted 19 June 2025
2347-5625/© 2025 The Authors. Published by Elsevier Inc. on behalf of Asian Oncology Nursing Society. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

just 33%.[12] In the UK, population-based targeted screening in Liverpool and Manchester reported participation rates of 40% and 26%, respectively.[13,14] These rates significantly lower than the 70% target recommended by the WHO,[15] which poses a major challenge to the early detection of lung cancer. The factors contributing to the low rate of LCS uptake are multifaceted, including a lack of awareness regarding LCS, a lack of health insurance, concerns about radiation exposure and potential future costs, poor access to health care services, logistical challenges, and negative attitudes regarding potential outcomes.[16,17] To address these barriers and improve LCS uptake, various interventions have been developed.

A recent systematic review analysed various interventions aimed at improving lung cancer awareness, encouraging help-seeking behaviours, and promoting early detection across all age groups.[18] However, this review didn't perform a subgroup analysis, thus it remains unclear which intervention is most effective. The other review summarized education strategies aimed to increase lung cancer awareness and knowledge.[19] However, this review was limited to four English databases and only focused on one type of intervention. Another systematic review summarised the evidence on the effectiveness of digital interventions in promoting LCS among eligible participants.[20] However, this review only searched three English databases and focused on one type of intervention. Recently, a systematic review was conducted to explore the characteristics and efficacy of interventions in promoting LCS decision making and behaviors.[21] However, this review lacked a clear definition of high-risk populations, and the absence of predefined criteria for participant subsets resulted in extreme variability in the proportion of high-risk populations (1.9%–100%) across studies. This variability precluded subgroup analyses and may reduce the applicability of the results. Overall, existing evidence on the most effective interventions to improve LCS uptake is established, and findings are inconclusive. There are methodological concerns in current reviews, such as insufficient databases or focusing only on English databases. This review aims to comprehensively and systematically evaluate the interventions or combinations of interventions that are most effective in increasing LCS uptake among high-risk individuals, and provide recommendations on the optimal content and format.

Methods

Search strategies

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Review and Meta-analysis guidelines.[22] A systematic search was conducted across multiple electronic databases, namely PubMed, Web of Science, Scopus, Cochrane Library, Medline, Embase, China National Knowledge Infrastructure, and Wan Fang, covering all records published up to 15 June 2024. Additional relevant studies were identified through manual searches of the reference lists from the retrieved studies. A full list of comprehensive search strategy is provided in Appendix A.

Eligibility criteria

Studies published in English or Chinese were included. No restrictions were imposed on the year of publication or country of origin. The eligibility criteria were predetermined using the PICOS framework.[22] The criteria were as follows. (1) Participants: in most guidelines, age and smoking history are the main criteria to define high-risk individuals.[8–10] However, air pollution, environmental exposures, and genetic factors also contribute to approximately 20% of lung cancer cases.[23] To ensure that individuals with a high risk of lung cancer, including nonsmokers, are not overlooked, our review defines high-risk individuals as adults without a history of lung cancer who meet at least one of the following criteria: a) age between 50 and 80 years; b) a smoking history of $\geq$ 20 pack-years, including those who currently smoke or have quit smoking within the past 15 years; c) passive smoking exposure, defined as living with or working in close proximity to a smoker for over 20 years; d) a family history of lung cancer; e) have chronic obstructive pulmonary disease; and f) occupational exposure to asbestos, rubber, dust, pesticides, radiation, beryllium, uranium, or radon for at least 1 year. Studies were excluded if the majority of the participants were not high-risk individuals or if the study did not conduct a subgroup analysis focusing on high-risk individuals.[24] (2) Intervention: any single or combined interventions, campaigns, programmes, trials, or educational efforts aimed at increasing LCS uptake were included. (3) Control: any type of comparison group, such as standard care or no intervention, were included. (4) Outcomes: including LCS uptake as primary outcome, which encompassed the use of LDCT, chest X-ray, sputum cytology; secondary outcomes including LCS intention, LCS knowledge, and decision conflict. (5) Study design: RCTs. Studies were excluded if they were grey literature, duplicate articles, protocols, reviews, editorials, commentaries, research letters, conference abstracts, or dissertations/theses or lacked full-text availability.

Study selection

After duplicates were removed using EndNote 20, two reviewers (YY and SZ) screened the titles and abstracts for eligibility, excluding irrelevant records. The full texts of potentially eligible studies were then reviewed, and the reasons for exclusion were recorded by both reviewers (YY and SZ). A manual search of the reference lists from these studies was also performed to identify additional relevant studies. Any conflicts during the screening process were resolved by a third reviewer (WZ).

Data extraction and synthesis

Data were extracted using a self-developed data extraction table that included details such as author(s), year, country, study design, theoretical framework, sample size, setting, relevant outcomes, intervention, control condition, follow-up duration, and findings. Two reviewers (YY and SZ) independently extracted the data. Subsequently, the extracted data were cross-checked and finalised with those obtained by a third reviewer (WZ). A meta-analysis was performed for the same outcome reported in at least two included studies by using RevMan 5.4.[25] For multi-arm studies, relevant experimental intervention groups were combined into a single group.[26] The effect sizes of dichotomous outcomes were pooled using odds ratios (ORs) with 95% confidence intervals (CIs), and for continuous outcomes, standardized mean differences (SMDs) were used. All meta-analyses were conducted using the random-effects method, and a P value of < 0.05 was considered statistically significant. Heterogeneity between studies was assessed using the I^2 statistic, categorised as low ($I^2 < 25\%$), medium ($25 \leq I^2 < 50\%$), and high ($I^2 \geq 50\%$).[27] Subgroup analyses were conducted to explore the effects of different intervention categories on LCS uptake, and sensitivity analysis was used to test the stability of the results by leave-one-study-out. Publication bias in the presence of a sufficient number of included studies ($n > 10$) was assessed through the funnel plots.[28]

Risk of bias assessment

The risk of bias in the retrieved studies was independently assessed by two reviewers (YY and SZ). Any disagreements were resolved through discussion with a third reviewer (WZ) to reach a consensus. The Cochrane Risk of Bias 2.0 tool for RCTs was used to evaluate the risk of bias in terms of the randomisation process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result.[29] Each domain was rated as high risk, low risk, or some concerns based on an algorithm considering several signaling questions. Overall risk of bias was categorized as: low risk if all

key domains were low risk; high risk if at least one domain was high risk; or some concerns if one or more domains raised concerns but no domain was high risk.

Confidence in cumulative evidence

Grades of Recommendation, Assessment, Development and Evaluation evidence grading system was used in assessing the overall quality of evidence generated from meta-analysis.[30] Five domains including methodological quality, directness of evidence, heterogeneity, precision of effect estimates, and risk of publication bias, were assessed. The quality of evidence was ranked with four levels, including high, moderate, low, and very low.

Results

Search results

A total of 7465 studies were identified, and 2606 duplicates were removed. The titles and abstracts of 4859 studies were screened, resulting in the exclusion of 4792 studies that did not meet the inclusion criteria. The full texts of 67 studies were retrieved for further screening based on the selection criteria, and 55 studies were excluded. Finally, the inclusion of two pilot RCTs reported in a single study (Kathuria et al., 2022) resulted in a total of 13 trials analyzed (Fig. 1), for clarity, 1 Kathuria et al. (2022) denotes Pilot 1 and 2 Kathuria et al. (2022) denotes Pilot 2 as described in the original publication.[31–42]

Risk of bias assessment

Fig. 2 presents the assessment results for the five domains and the overall rating of bias for individual trials. Overall, seven trials were identified as having some concerns regarding the randomisation process, deviations from intended interventions, and selection of reported results.[31,33,34,36,39,40] Four studies were assessed as having a low risk of bias.[32,35,37,41] Two trials were considered to have a high risk of bias concerning the selection of reported results.[38,42]

Characteristics of participants and settings

The characteristics of the included studies are summarised in Table 1. All 13 trials were RCTs published between 2017 and 2024. Ten studies were conducted in the USA, and only two studies were conducted in the UK.[35,37] Nearly all participants were smokers aged 50 years or older who had a high risk of lung cancer. The sample sizes ranged from 21 to 2347. One trial was conducted in the community,[37] the majority (11/13) in health care settings, and one trial in tobacco quit lines.[42]

Theoretical framework

Only three trials adopted theoretical frameworks to guide the development of their interventions. These frameworks included the Preventive Health Model ($n = 1$),[36] the health belief model ($n = 1$),[33] and a combination of the theory of planned behaviour and self-determination theory ($n = 1$).[32]

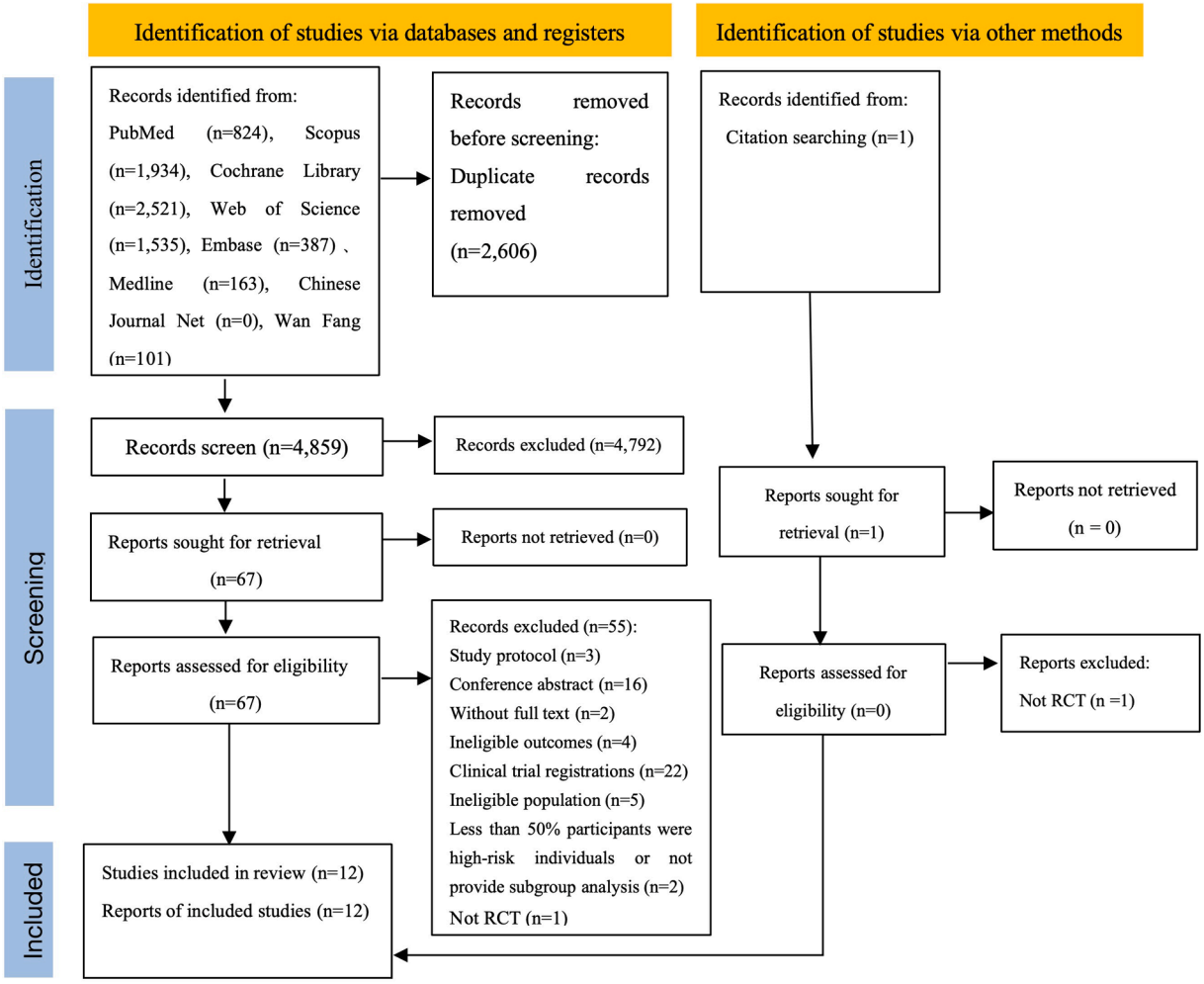


Fig. 1. Flow diagram of the studies selection process.

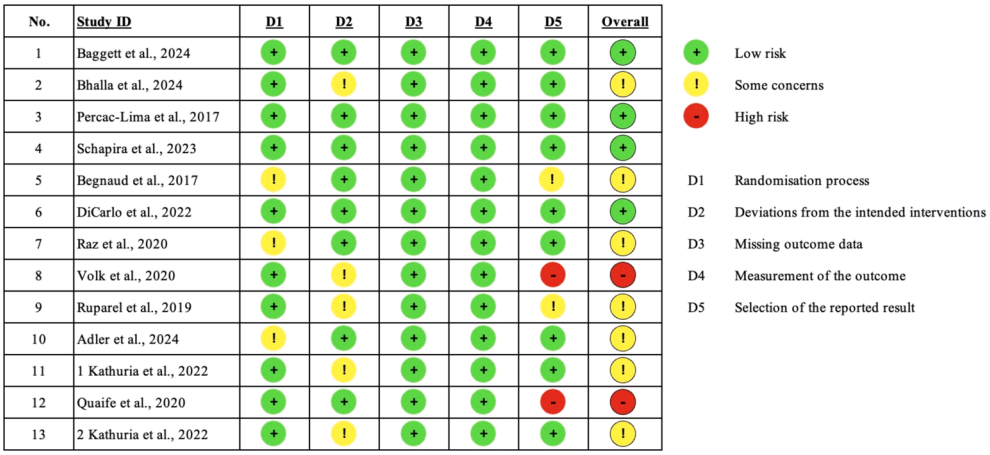


Fig. 2. Risk of bias of included studies.

Characteristics of interventions

(1) Intervention Components. Six type of interventions were identified, including online education,[39,40] shared decision-making,[35,41,42] patient navigation,[32,34,37] outreach promotion,[33,35,40] targeted invitation materials,[38] and combined interventions.[34,35] The online education mainly disseminated information on lung cancer and LCS, eligibility, expectations, the benefits and harms of screening, LDCT procedure, possible results after the scan, and decision-making assistance. Shared decision-making included an overview of LCS using simulated patient–clinician dialogues, interactive knowledge related to LCS, and a pictograph representing screening outcomes. Patient navigation not only provided educational information but also explored participants' concerns regarding and barriers to screening, scheduled appointments with primary care physicians and LCS services, and helped overcome barriers. Outreach promotion involved making follow-up contact with participants through telephone, mail, or messaging after the initial contact to assess LCS eligibility, obtain consent, and discuss screening options. Targeted invitation materials included a stepped invitation strategy, starting with a low-burden leaflet that provided sufficient information for individuals to decide whether to attend and consider the screening offer. Full information was then provided either at the appointment using the control group's booklet or made available beforehand through a website, phone, or post. Two trials conducted multicomponent interventions; one study combined outreach promotion with decision counselling using an online interactive decision support application,[35] whereas another trial used inpatient shared decision-making combined with community health worker–led navigation.[36] (2) Intervention provider. Three trials were delivered by navigators who had undergone focused education and training in patient navigation, LCS, motivational interviewing, problem-solving, and goal setting. Five trials were delivered by primary care providers who had received specific training in the consent process or by research staff, and five trials did not report the intervention provider. (3) Intervention format and delivery mode. Two trials used face-to-face method to conduct the interventions, two trails combined face-to-face and online methods, whereas online methods were the mostly used format, such as 5.5- to 30-min videos, telephone calls, mail or messages, and software or tools. For delivery modes, nearly all trials delivered individually, only one trail was delivered in the group format.[38] (4) Intervention dosage. Patient-directed dosage ($n = 6$) was widely used among the included trials. Besides, three trials used a single session, one trials sent six text messages at 15-day intervals, whereas other three trials did not provide detailed information on the dosage. The follow-up duration ranged from 1 week to 9 months.

Characteristics of comparators

Eight trials used usual care as the comparator, such as a nontailored lung screening information sheet. Two trials reported in one study used enhanced usual care.[36] Two trials reported no intervention in the control group.[33,39] One study used an attention control intervention titled 'Cancer Prevention: An Information Guide' to reduce threats to internal validity.[41]

Effects of interventions on LCS related outcomes

LCS uptake

All included trials reported the effects of various interventions on LCS uptake that were either self-reported by participants or obtained from medical records. The results indicate that these interventions improved LCS uptake (OR, 1.70; 95% CI, 1.10–2.62; $P = 0.02$; $I^2 = 87\%$) compared with no intervention or usual care (Fig. 3). Subgroup analysis revealed that patient navigation (OR, 3.31; 95% CI, 1.70–6.41; $P < 0.001$; $I^2 = 79\%$), and combined interventions (OR, 3.38; 95% CI, 2.08–5.52; $P < 0.001$; $I^2 = 0\%$) significantly improved LCS uptake rates. In contrast, online education (OR, 0.86; 95% CI, 0.54–1.35; $P = 0.51$; $I^2 = 0\%$), shared decision-making (OR, 1.31; 95% CI, 0.54–3.14; $P = 0.55$; $I^2 = 68\%$), and outreach promotion strategies (OR, 1.13; 95% CI, 0.14–8.97; $P = 0.91$; $I^2 = 68\%$) did not exert a significant effect (Fig. 4). Besides, a sensitivity analysis confirmed that the effect size remained consistent when each study was removed from the analysis. As more than 10 studies were included, a funnel plot was constructed (Fig. 5). It appeared largely symmetrical, suggesting a low risk of publication bias.

LCS intention

Two trials measured intention for LCS, and both showed that there were no significant difference in intentions to be screened between the intervention group and the control group.[38,42]

LCS knowledge

Four trials assessed LCS knowledge, with three included in the meta-analysis using different scales, while one was excluded due to being a binary variable rather than a continuous variable. And the meta analysis showed that these interventions improved LCS knowledge (SMD, 1.97;

Table 1
Characteristics of included trials ($N = 13$).

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
Baggett et al., 2024 USA	2-arm RCT Boston Health care for the homeless program Health belief model	Those aged 55–77 years, at least a 30 pack-year smoking history, and smoking within the past 15 years Recruited through a combination of in-person and phone-based outreach as well as referrals from primary care practitioners	IG: $n = 173$ CG: $n = 87$ IG: 60.5 ± 4.4 y CG: 60.7 ± 5.4 y Male: 184/260 (70.8%)	IG: Patient navigation ✓ Providing lung cancer education on the benefits and risks of LCS ✓ Facilitating LCS shared decision-making visits with PCPs ✓ Assisting in scheduling LDCT appointment ✓ Facilitating LDCT appointment attendance through reminder phone calls and transportation assistance ✓ Arranging follow-up studies or office visits when needed ✓ Offering tobacco cessation support for current smokers, and coordinating care with primary care practitioners, referral specialists, and other care team members. CG: Usual care ✓ Basic educational materials on general lung health and referred back to their primary care practitioners to discuss whether to pursue LCS per usual practice	Navigator were college graduates who underwent focused education and training in patient navigation, LCS, and tobacco counseling	Patient-directed Individual Phone call/ In person	LDCT uptake (obtained from medical records)	6 months	• 75 of the 173 participants in the IG (43.4%) and 8 of the 87 participants in the CG (9.2%) had completed LCS LDCT ($P < 0.001$)
Bhalla et al., 2024 USA	2-arm RCT Parkland health –	Those aged 55–77 years, 30 pack-years smoking history, and had quit smoking within the past 15 years if a former smoker Contacted by telephone based on electronic health record	IG: $n = 225$ CG: $n = 222$ IG: 62.6 ± 4.6 y CG: 61.4 ± 4.7 y Male: 243/447 (54%)	IG: Patient navigation ✓ Intake: Introduce self and provide contact information; assess barriers to completing LCS; provide information and resources as needed; determine smoking status and interest in smoking cessation ✓ Reminder (2–3 days before a scheduled LDCT screening or other screening-related appointment): Remind patient about LDCT appointment; Identify potential barriers to attending appointment ✓ Follow-up (3–7 days after the scheduled appointment): Follow up with patient about LDCT appointment; If patient missed appointment, encourage them to reschedule; follow up on smoking status and interest in smoking cessation ✓ Outreach (triggered by subsequent screening or	Navigators were research assistants with a background in clinical outreach and engagement who were trained by using an abbreviated curriculum focused on motivational interviewing adapted from the george Washington university cancer institute. Navigators also received supplemental training on patient resources within the health care system and behavioral aspects of LCS	Patient-directed Individual Systematic phone call	LDCT uptake (obtained from electronic health record)	Not reported	• 86% patients in the IG completed the initial LDCT screen compared with 79% patients in the CG ($P = 0.03$)

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				diagnostic orders): Inquire about next LCS appointment and assess potential barriers; provide information and resources as needed ✓ Using motivational interviewing techniques to assess interest in smoking cessation and to empower patients to connect with parkland's smoking cessation resources CG: Usual care					
Percac-lima et al., 2017 USA	2-arm RCT Community health centers –	Current smokers aged 55–77 years Recruited through a electronic medical record	IG: $n = 400$ CG: $n = 800$ IG: 61.8 ± 5.4 y CG: 62.4 ± 5.7 y Male: 570/1200 (47.5%)	IG: Patient navigation ✓ Performing an initial interview with patients to determine smoking history/eligibility for LCS ✓ Reviewing the purpose of LCS, the risks of screening, such as exposure to radiation and false positive results, and the benefits ✓ Exploring their concerns and barriers to screening ✓ Arranging a shared decision-making appoint-ment with participants' primary care practitioners so LCS could be discussed and ordered, if appropriate if they were interested in screening. ✓ Reminding the patient about the scheduled CT screening test, helping with translation, insurance issues, transportation, and overcoming other system barriers as needed. ✓ Providing brief smoking cessation counseling. If interested in quitting, smokers were connected to existing smoking cessation resources. Place a referral to the Massachusetts quitline or to a tobacco cessation specialist working in the community health centers. If the patient was interested in smoking cessation medication, the navigator would notify the provider ✓ After a patient completed LCS, the navigator reviewed the CT results in the EMR. If the patient had an abnormal test result that required follow-up, the navigator contacted	Four part-time lay patient navigators with bachelor's degrees, trained in motivational interviewing, problem solving, goal setting, use of the IT system, and electronic medical record documentation, and educated about lung cancer and screening with LDCT.	Patient-directed Individual Phone call	LCS CT uptake (obtained from medical records)	–	• LDCT were performed more often in the intervention group compared to the control group (23.5% 94 vs. 8.6% 69, $P < 0.001$)

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				the patient's primary care practitioners and helped the patient obtain timely follow-up. ✓ The navigators used the population health management IT registry tool to follow patients randomized to the intervention group and to track lung CT tests scheduled and completed. They also recorded each patient contact and documented interventions performed. Actions taken at shared decision-making visits with the primary care practitioners were documented in the electronic health record. CG: Usual care ✓ Primary care practitioners in the usual care group could identify their eligible patients, discuss screening at a SDM visit and order a screening CT, but without support from navigators					
Schapira et al., 2023 USA	2-arm RCT Veteran affairs medical centers –	Those aged 55–80 years with a smoking history of at least 30 pack-years who were current smokers or had quit within the past 15 years Use used prescription data (varenicline tartrate or nicotine replacement therapy) or positive screen results for active smoking in primary care or dental clinic notes to identify potential participants	IG: $n = 69$ CG: $n = 71$ IG: 64 (61.0–69.0) y CG: 64 (61.8–69.5) y Male: 129/140 (92.1%)	IG: Shared decision making—LCSDecTool ✓ The LCSDecTool was designed to be used independently by the patient before the clinic visit with the option to share some components with the clinician during the clinic visit ✓ The tool includes an overview of LCS using a simulated patient-clinician dialogue, interactive knowledge boxes providing brief information on 6 key attributes related to LCS, a pictograph representing LCS outcomes, a value elicitation exercise, smoking cessation advice, mental health resources, and the option to request a referral to a smoking cessation clinic or to a behavioral health clinician to support smoking cessation efforts ✓ The tool includes a text field where a patient can enter questions for their clinician. A summary page is created by the LCSDec tool that displays	Not reported	Patient-directed Individual Online	LDCT uptake (obtained from electronic record review) LCS knowledge Decisional conflict	1 month 3, 6, 9 months	• LDCT uptake was greater in the IG at 6 months (26 of 69 [37.7%] vs 15 of 71 in the CG [21.1%]; $P = 0.04$); persisted at 9 months (31 of 69 [44.9%] vs 18 of 71 [25.4%]; $P = 0.02$) • Mean LCS knowledge score was greater in the IG immediately after intervention (7.0 [95% CI, 6.3–7.7] vs 4.9 [95% CI, 4.3–5.5]; $P < 0.001$) and remained higher at 1 month (6.3 [95% CI, 5.7–6.8] vs 5.2 [95% CI, 4.5–5.8]; $P = 0.03$) and 3 months (6.2 [95% CI, 5.6–6.8] vs 5.1 [95% CI, 4.4–5.8]; $P = 0.01$). • The IG's mean decisional conflict score decreased immediately after intervention compared with the CG, but this effect did not persist at 1 month and 3 months.

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				participant value ratings and questions entered for the clinician. From the opening page, a clinician can access a clinician portal that directly connects to the pictograph of LCS outcomes and the summary page. CG: Web-based attention control intervention entitled cancer prevention: An information Guide. ✓ The 10-page guide provides general information on cancer prevention and the USPSTF screening guidelines for breast, colon, cervical, and lung cancer					
DiCarlo et al., 2022 USA	3-arm RCT Jefferson health Preventive health model	Those aged 50–80 years of age and at high risk for lung cancer Recruited from electronic medical record	OC-DC: $n = 302$ OC: $n = 297$ CG: $n = 1748$ – Male: 1107/2347 (47.2%)	OC-DC: Decision counseling + outreach contact ✓ Mailed a cover letter that described the study and provided contact information for opting out of future study contacts, and LCS educational materials—an informational booklet (providing information on LCS eligibility, LDCT screening test, steps in the screening process, and the possibility of lung nodule detection) and a print decision aid (presenting answers to the following questions: "What does the screening test involve"; "what does the screening test look for?"; "what is my chance of having lung cancer diagnosed?"; "what are the benefits?"; and, "what are the risks?"). ✓ The care coordinator made telephone contact within 10 days after the mailing, reviewed the mailed intervention materials and used an online interactive decision support software application (the decision counseling Program© or DCP) to guide participants through a brief (~10 min) decision counseling session focused on eliciting values and clarifying preference related to LCS. (The score provides a measure of personal preference (i.e.,	Care coordinator from the practice	Patient-directed Individual Emile, phone call	LDCT uptake (obtained from electronic record review)	Immediate 6 months 8 months	• LCS uptake was higher in the OC-DC than in the CG (21/302 vs 31/1748); LCS uptake was higher in the OC than in the CG (12/297 vs 31/1748).

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				to be screened, unsure about being screened, not to be screened) ✓ At the end of the session, the care coordinator used the application to compute an LCS preference score. The care coordinator validated the results with each participant. ✓ Finally, the care coordinator offered to schedule a JLCSP appointment for the participants and encouraged the participant to discuss LCS with their primary care physician at a future office visit. OC: Outreach contact ✓ Mailed a cover letter that described the study and provided contact information for opting out of future study contacts, and LCS educational materials—an informational booklet (providing information on LCS eligibility, LDCT screening test, steps in the screening process, and the possibility of lung nodule detection) and a print decision aid (presenting answers to the following questions: "What does the screening test involve"; "what does the screening test look for?"; "what is my chance of having lung cancer diagnosed?"; "what are the benefits?"; and, "what are the risks?"). ✓ The care coordinator made telephone contact within 10 days after the mailing, reviewed the mailed intervention materials, assessed interest in screening, and offered to schedule a JLCSP appointment for those who expressed interest, and encouraged participants to discuss LCS with their primary care physician at a future office visit CG: Usual care					

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				✓ Without any intervention contacts					
Raz et al., 2020 USA	2-arm RCT Kaiser permanente southern California –	Smokers aged 55–80 who underwent smoking cessation services with at least 30 pack-years smoking history The study population included patients who completed at least 1 in-person smoking cessation group session, attended a telephone Wellness coaching visit, or viewed an online video for quitting Smoking between January 1, 2016, and February 2018	IG: $n = 1026$ CG: $n = 255$ IG: 64.0 ± 6.1 y CG: 63.5 ± 5.5 y Male: 508/1026 (49.5%)	IG: Online education video titled "LCS: Options" ✓ Including information on lung cancer, LCS, eligibility, what to expect, and decision-making assistance. The video also contains instructions about how participants can obtain LDCT for LCS. This video was developed by emmi solutions utilizing evidence-based information about LCS and was developed with patient input to reflect the patient experience and communicate information effectively to a lay audience. Graphics were developed by professional medical illustrators ✓ At the end of the LCS education video, patients were asked to complete a survey that included questions about smoking habits, knowledge of LCS and likelihood of getting screened for lung cancer, and history of utilization of other cancer screening testing including colorectal cancer screening and mammography. ✓ Whether a portionor all of the video was watched, and how much of the video was watched can be identified by research team, utilizing analytics built into the educational software. CG: No intervention	Not reported	30 minutes online educational video Individual Online	LDCT uptake	6 months	• 37 (3.6%) underwent LDCT in the IG compared to 11 (4.3%) in the CG during the 6-month follow-up period ($P = 0.59$).
Volk et al., 2020 USA	2-arm RCT Tobacco quit lines –	Quit line clients (ages 55–77 years) who reported a 30-plus pack-year smoking history Quit line clients	IG: $n = 259$ CG: $n = 257$ – Male: 196/516 (38%)	IG: Decision aid ✓The PDA was a 9.5-min narrated video, LCS: Is it right for me? (Details of the development process have been previously described). The development process followed the standards of the international patient decision aid standards collaboration, and the PDA met national quality forum certification criteria for PDAs. ✓The narrated PDA included information about (1) eligibility for LCS and a calculation of tobacco pack-	Not reported	9.5-min narrated video Individual Online	LDCT uptake (self reported) Decision conflict LCS knowledge LCS intention	Baseline 1-week 3-month 6-month	• By the 6-month follow-up, 57 participants in the IG completed compared with 68 in the CG. • IG participants scored lower (better) on the informed and values clarity subscales of the decisional conflict scale than did CG participants. Using cut points for clinical significance • Knowledge was significantly higher in the IG than those in the CG at each follow-up assessment period

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				year smoking history, (2) lung cancer epidemiology and risk factors, (3) a video of a patient in a CT scanner, (4) icon arrays to graphically depict the magnitude of mortality reduction, false-positive results, and harms from invasive diagnostic procedures, and (5) radiation exposure depicted within the context of other sources of radiation (e.g., a screening mammogram). ✓The development process followed the standards of the international patient decision aid standards collaboration, 21 and the PDA met national quality forum certification criteria for PDAs. We updated our PDA to reflect the US preventive services task force 2014 recommendation and eligibility criteria for LCS from the CMS. CG: Usual care ✓Standard educational materials, a 2-page brochure from a lung cancer advocacy group and included structured questions a patient can ask a physician about LCS. The questions addressed (1) eligibility for screening, (2) the harms and benefits of screening, (3) what to expect from undergoing an LDCT scan, (4) the costs of screening, (5) how to interpret the LDCT results, (6) the importance of smoking cessation, and (7) where to find more information about lung cancer and screening.					• No group differences in intentions to be screened (70.8% [165 of 233] in the IG, and 65.1% [151 of 232] in the CG)
Ruparel et al., 2019 UK	2-arm RCT Local London hospital –	Smokers and former smokers (within 5 years of quitting) 60–75 years of age Identified from primary care records	IG: $n = 126$ CG: $n = 120$ – Male: 109/246 (44.3%)	IC: Online education ✓ An information film and given the same information booklet (available at https://www.roycastle.org/lungcancerscreeningguide). ✓ Both interventions discussed lung cancer, the benefits and harms of LCS (including indeterminate pulmonary nodules and false positives, overdiagnosis, and radiation damage), the LDCT procedure, and the possible results after the scan.	Health care professionals, were nurses or clinical trial practitioners who had been specifically trained in the consent process	A 5.5 minutes information film Group Face-to-face	LDCT uptake (self reported) Decision conflict LCS knowledge	Post intervention	• LDCT uptake did not significantly differ across groups ($P = 0.66$), with 76.7% and 78.9% proceeding to LDCT in the IG and the CG • The IG had lower decision conflict (mean 8.5/9 points [SD 1.3) than the CG (mean 8.2/9 points [SD 1.5]). • Mean objective knowledge scores increased by 2.16 (SD 1.8) and 1.84 (SD 1.9) in the IG and CG.

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				✓The booklet was 10 pages long and designed to be clear and comprehensible for individuals with a reading age of 11–13 years. CG: Usual care ✓ Information booklet used for LSUT's control invitation materials. The booklet was 10 pages long and designed to be clear and comprehensible for individuals with a reading age of 11–13 years. Both interventions discussed lung cancer, the benefits and harms of LCS (including indeterminate pulmonary nodules and false positives, overdiagnosis, and radiation damage), the LDCT procedure, and the possible results after the scan.					
Adler et al., 2024 USA	2-arm pilot RCT Emergency department Theory of planned behavior, Self-determination theory	Adults aged 50 to 80 requires ≥ 20 pack-year history and either currently smoking or having Quit within the past 15 years Electronic ED tracking board	IG: $n = 103$ CG: $n = 95$ IG: 59.6 ± 7.3 y CG: 59.8 ± 7.8 y Male: 109/198 (55.1%)	IC: Outreach contact ✓A basic referral as the control group. ✓A series of behavioral text-messages. Each message had two elements. ✓The first element of each message included a statement directed toward one of the theory of planned behavior predictors of behavioral intentions or one of the self-determination theory promoters of autonomous motivation. This behavioral element of the message was tailored depending on whether the participant had ever been screened before. ✓ The second element of each message was a reminder to schedule LCS with their primary care practitioners or the university of rochester LCS Program's patient navigator. To facilitate patient follow-through, hyperlinks to the relevant telephone numbers of the PCP (if they had one) and the LCS patient navigator were included in each message. CG: Usual care ✓ Informed of their eligibility for LCS and received eligibility for LCS and	Research staff	Six text messages, at 15-day intervals Individual Online	LDCT uptake (obtained from electronic record review)	5 months	• LDCT rates at follow-up were similar across the IC and the CG (8/103 vs 14/95, $P = 0.119$) • The IG intervention significantly improved normative perceptions of LCS (an element of TPB) compared to the CG (mean 4.74 [SD 1.65] vs 5.37 [SD 0.97], $P = 0.015$)

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				received a basic referral consisting of a verbal recommendation and written instructions to consult with their primary care practitioners about LCS (if they had one) and given the contact information to the university of rochester LCS program patient navigator					
Kathuria et al., 2022 USA	Two 2-arm pilot RCT Boston Medical center –	Those aged 55–80 years, ≥ 30-pack years smoking Identified from tobacco treatment consult service	Pilot RCT 1 IG1: $n = 52$ CG1: $n = 48$ IG1: 62.1 ± 5.0 y CG1: 61.6 ± 5.8 y Male: 65/100 (65.0%) Pilot RCT 2 IG2: $n = 10$ CG2: $n = 11$ IG2: 64.0 ± 6.1 y CG2: 64.0 ± 5.6 y Male: 15/21 (71.4%)	Pilot RCT 1 IG1: Shared decision making ✓ Conducting a tailored discussion on tradeoffs of CT screening using a decision aid ✓ Connect interested patients to CT screening (nurse practitioner notifies primary care practitioners of patients desiring LCS through the electronic health record, encouraging primary care practitioners to place referral) CG1: Enhanced usual care ✓ The nurse practitioner gave participants the LCS patient decision aid but did not review it together, and instructed participants to let their primary care practitioners know if they were interested in LCS Pilot RCT 2 IG2: Inpatient shared decision making + community health worker led-navigation ✓In the adapted intervention, the community health workers screened for unmet social needs that are barriers to LCS by asking if participants have (1) trouble getting transportation to medical appointments, (2) difficulty with affordability of copayments, and (3) access to a working cell phone to be reminded of patient appointments. The community health worker followed participants for 3 months or until LCS completion (whichever came first), assisting with scheduling LCS and transportation, and arranging subsidized phone services as needed. CG2: Enhanced usual care	Nurse practitioner	Pilot RCT 1 One session Individual Face-to-face Pilot RCT 2 Patient-directed Individual Face-to-face & online methods	LDCT uptake	3 months	Pilot RCT 1 • No difference between study arms [IG1, 4% (2/52); CG1, 2% (1/48), $P = 0.6$]. Pilot RCT 2 • LCS completion was higher in the intervention arm [IG2, 50% (5/10); CG2, 9% (1/11), $P < 0.05$]

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				✓ The nurse practitioner gave participants the LCS patient decision aid but did not review it together, and instructed participants to let their primary care practitioners know if they were interested in LCS					
Quaife et al., 2020 UK	2-arm pilot RCT Hospital outpatient clinics –	Those aged 60–75, who recorded as a current smoker within the last seven years, no pre-specified exclusion criteria contraindicating LDCT screening	IG: $n = 1006$ CG: $n = 1006$ IG: 66.1 ± 4.3 y CG: 65.9 ± 4.3 y Male: 931/2012 (53.7%)	IG: Targeted invitation materials ✓ An 'M.O.T. for your lungs' leaflet, designed to target psychological barriers to attendance (fear, fatalism and stigma), to be low burden (sufficient for deciding to attend and consider the screening offer) and stepped (full information given at the appointment using the control group's booklet, or available before via a website, phone or post). An 'M.O.T.' is an annual roadworthy test for vehicles and was a lay concept perceived to be analogous to a medical check-up preferred by patient and public involvement groups ✓ Received the same postal invitation letters from their primary care practice: Pre-invitation letter, invitation letter with scheduled appointment, and reminder re-invitation letter with a second scheduled appointment (sent to non-responders > 4 weeks after missed appointment). The letters were identical with two exceptions: 1) the intervention group's letters referred to 'ever smokers' whereas the control group's referred to 'current and former smokers', and 2) the intervention group's invitation letter included a bullet-pointed summary of the lung health check, including LDCT scan offer, on the reverse side. CG: Usual care ✓ Information booklet mimicking 'the facts' booklets of NHS cancer screening programmes	Not reported	No information Individual Email	LDCT uptake LCS intention LCS knowledge Decision conflict		• No difference between IG (52.3%) and CG (52.9%) in unadjusted (OR: 0.98, 0.82–1.16) or adjusted (OR: 0.98, 0.82–1.17) analyses • IG did not affect the intention of LDCT uptake. • There was no difference in mean scores for conceptual and numerical knowledge of the IG group [5.7 (2.3) vs 5.5 (2.3) of the CG] • There were no significant difference in decision conflict score between the two groups

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				✓ Received the same postal invitation letters from their primary care practice: Pre-invitation letter, invitation letter with scheduled appointment, and reminder re-invitation letter with a second scheduled appointment (sent to non-responders > 4 weeks after missed appointment). The letters were identical with two exceptions: 1) the intervention group's letters referred to 'ever smokers' whereas the control group's referred to 'current and former smokers', and 2) the intervention group's invitation letter included a bullet-pointed summary of the lung health check, including LDCT scan offer, on the reverse side.					
Begnaud et al., 2017 USA	2-arm pilot RCT Primary care or pulmonary clinics –	Patients aged 55–79 years who were former smokers Identified from electronic health record	IG: $n = 99$ CG: $n = 101$ IG: 62.0 y CG: 62.0 y Male: 109/200 (54.5%)	IG: Electronic promotion of LCS ✓ Participants were sent messages in the patient portal that informed them of their potential eligibility for a new preventive health care service. The message included a questionnaire that was linked to their smoking history in the electronic health record, which could be reviewed and modified by the patients. The letter instructed recipients to determine the number of years of smoking on the basis of the age at which daily smoking started and stopped, and to estimate the number of packs of cigarettes smoked each day for the time they were a smoker. To avoid influencing responses, the letter did not include eligibility criteria for LCS. ✓ All respondents were sent a follow-up portal message to indicate eligibility for LCS and to inform them of the risks and benefits of screening. If eligible, an order for LCS was placed in the chart and the patient was provided imaging scheduling information.	Not reported	No information Individual Message	LDCT uptake	6 months	• No difference between IG (4/99) and control (1/101) CG.

(*continued on next page*)

Table 1 (*continued*)

Studies & Country	Design & Setting &Theory	Participants	Sample & Age & Gender	Main Component	Interventionist	Dosage & Format & Delivery mode	Outcomes & Measurements	Follow-up time(s)	Findings*
				✓ If a patient failed to read the message after 30 days, a duplicate electronic message was sent. Patients who failed to read the second message were sent a letter via the US postal service with the same information. CG: No intervention					

LCS, lung cancer screening; IG, intervention group; CG, control group; LDCT, low-dose computed tomography; OC-DC, outreach contact plus decision counseling; OC, outreach contact; RCT, randomized controlled trial.

95% CI, 0.35–3.59; $P = 0.02$; $I^2 = 99\%$) compared with no intervention or usual care (Fig. 6).

Decision conflict

In evaluating decision conflict, four trials employed various scales. One trail using 10-item Decision Conflict Scale found the intervention group's mean decisional conflict score decreased immediately after intervention compared with the control group, but this effect did not persist at 1 month and 3 months.[41] One trial used the Informed and Values Clarity subscales of the Decisional Conflict Scale, showing that the intervention group scored lower (better) than the control group on both subscales.[42] Two trials used an adapted low-literacy Decision Conflict Scale. One found that the intervention group had lower decision conflict than the control group,[39] while the other showed no significant difference between the two groups.[37]

Quality of evidence

The quality of the studies included in this review was evaluated using the GRADE system. The quality of evidence for the LCS uptake and knowledge outcome were downgraded by two levels due to the high risk of bias and inconsistency of a high heterogeneity (Fig. 7).

Discussion

To the best of our knowledge, this is the first systematic review and meta-analysis of RCTs to investigate the effectiveness of various interventions in improving LCS uptake among high-risk individuals. All of the included studies were conducted in developed countries where LCS guidelines are well-established and screening procedures are standardised. However, given the different national conditions, policies, limited medical access, and high smoking rates in developing countries,[43] there is a critical need for these regions to focus more on LCS to help reduce the global burden of the disease.

Main findings

In the present review, the majority (9/12) of the included studies did not report using theoretical frameworks for designing their interventions, which may limit their effectiveness. Theoretical models can provide valuable guidance for developing interventions aimed at behaviour changes.[44] Thus, future studies should use theoretical models to develop and design interventions aimed at improving LCS uptake.

The present review provides strong evidence supporting the effectiveness of interventions aimed at increasing LCS. Subgroup analysis results revealed that patient navigation was effective in improving LCS uptake. Key components of patient navigation included identifying barriers to screening, providing necessary information and resources, reminding patients about LDCT appointments, and conducting follow-up calls. These elements substantially enhanced screening uptake among high-risk individuals. This finding is consistent with that of a previous systematic review that found patient navigation to be effective in increasing screening uptake for colorectal, breast, and cervical cancers.[44] The success of patient navigation is likely to result from its flexibility in providing multidimensional, personalised services throughout the cancer screening process.

Our review found that combined interventions were effective in improving LCS uptake. This is consistent with that of a previous systematic review indicating that multicomponent interventions effectively increased breast, cervical, and colorectal cancer screening in rural populations.[45] These combined interventions typically included outreach promotion, shared decision-making, and patient navigation. Outreach promotion raises awareness and generates interest, shared decision-making ensures informed and personalised choices, and patient navigation provides practical and emotional support. When these components are integrated, they create a comprehensive approach that

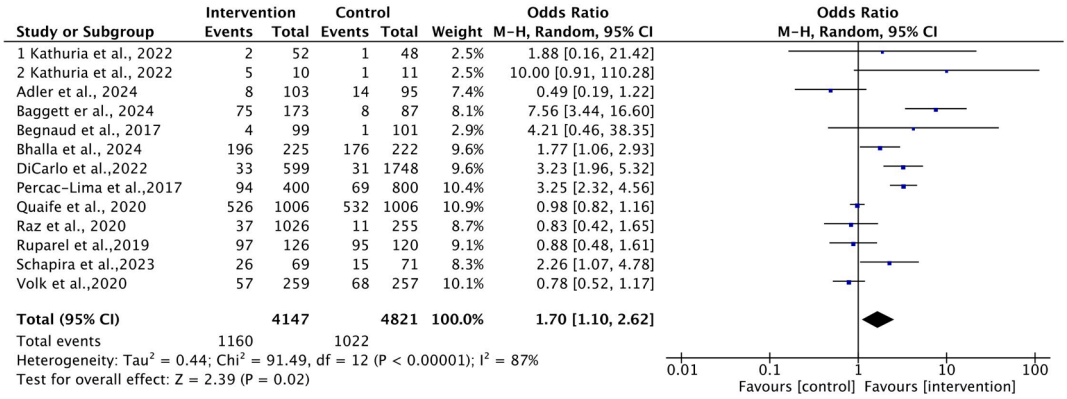


Fig. 3. Effects of various interventions on lung cancer screening uptake. CI, confidence interval.

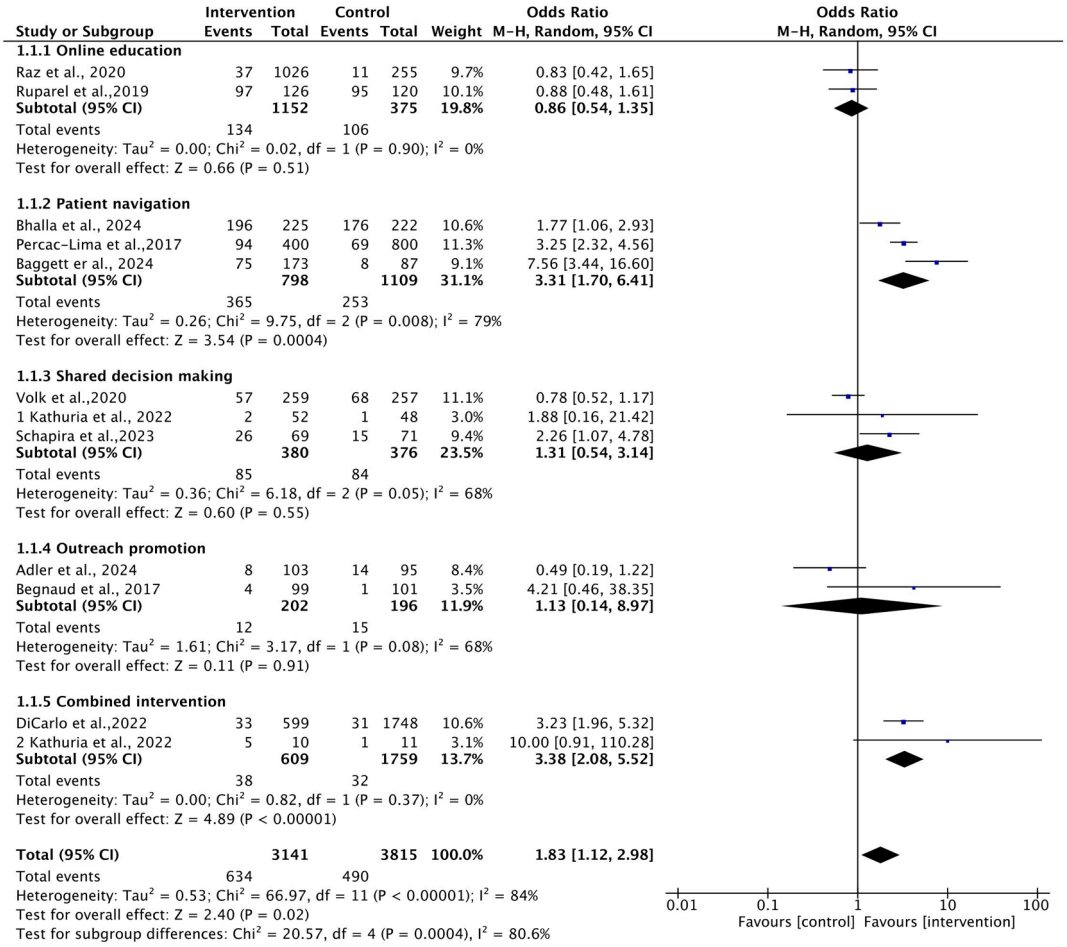


Fig. 4. Effects of intervention components on lung cancer screening uptake. CI, confidence interval.

addresses multiple aspects of the screening process, making it a highly effective strategy for increasing LCS uptake.

This review indicated that online education had no significant effect on LCS uptake, which is consistent with that of a previous systematic review that examined the effects of educational messages and strategies on LCS uptake.[19] Although education is recommended to improve cancer awareness, identify barriers, and facilitate health-seeking behaviour,[46] increasing knowledge alone may not be sufficient to change screening behaviours, especially for individuals facing complex barriers to screening. Thus, it is crucial to combine educational efforts with other intervention strategies in clinical practice. Our review found that outreach promotion did not significantly affect LCS uptake, a result that contrasts with the finding of a previous study.[47] This discrepancy may be due to the weak effect of mail or message-based interventions and inadequate education on LCS. Future efforts should focus on improving targeted information, expanding communication platforms, and presenting information clearly and comprehensibly to enhance outreach effectiveness.

Our review revealed that shared decision-making did not significantly improve LCS uptake, which is consistent with a previous systematic review that provided limited evidence that decision aid tools effectively change behaviour.[20] This lack of impact may be attributed to the low effectiveness of online interventions because most decision aids in our review were delivered through online tools, software, and videos.

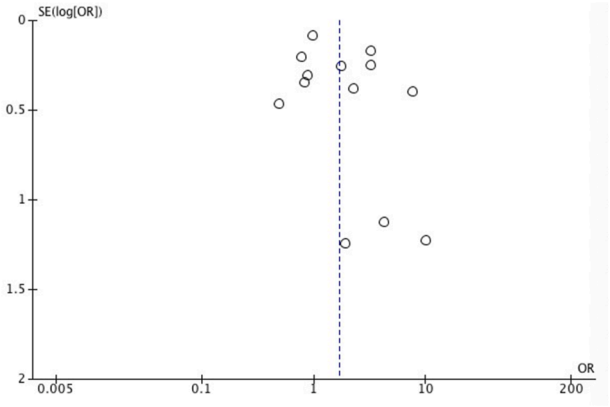


Fig. 5. Funnel plot of promoting interventions on Lung Cancer Screening Uptake.

In addition, some studies included in our review described the adverse effects of screening, including radiation exposure, false positives, and complications from follow-up procedures, which may have further reduced screening motivation.[48] However, some included studies suggest that shared decision-making may be effective in encouraging screening behaviors among high-risk individuals. Given the varying effects observed, the overall conclusions remain inconclusive. Future studies should investigate the effectiveness and feasibility of decision aids in LCS to provide clearer insights.

Strengths and limitations

This systematic review addresses a critical knowledge gap by evaluating the effectiveness of various interventions in increasing LCS uptake among high-risk individuals. A key strength of the present review is the inclusion of only RCTs that evaluated the effectiveness of interventions in improving LCS uptake among high-risk individuals. This focus on RCTs enhances the quality and reliability of our findings. However, this review has several limitations that should be considered. First, most of the included studies had some concerns regarding the risk of bias, particularly due to inadequate allocation concealment or missing outcome data, and two studies were identified as having a high risk of bias. Second, all included studies were conducted in the USA or the UK, potentially limiting the generalisability of the findings to other countries. A further limitation of this review is the inconsistent reporting of intervention dosages, often due to the patient-centered design of the studies. This variability limits the development of standardized dosage guidelines and the applicability of findings to clinical practice. Additionally, there was high heterogeneity among studies due to different intervention designs. Despite conducting sensitivity and subgroup analyses, the results should be interpreted with caution. Finally, the review only included publications written in English and Chinese. Thus, some relevant studies published in other languages may have been missed.

Implications for nursing practice and research

The review findings indicate that patient navigation and combined strategies—such as outreach promotion and shared decision-making—are effective in promoting early detection, particularly among smokers. However, as around 20% of lung cancer cases occur in non-smokers, future research should also target high-risk nonsmoking populations. Future studies should employ theory-based randomized controlled trials with rigorous designs, including proper randomization, allocation concealment, and handling of missing data. As all included studies were conducted in the U.S. or U.K., evaluating these interventions in diverse cultural and health care settings is essential for broader applicability. Incorporating these strategies into clinical practice may enhance early detection and improve outcomes.

Conclusions

This systematic review thoroughly examined the effectiveness of various interventions designed to improve LCS uptake. Our findings suggest that patient navigation and combined interventions are particularly effective because they offer tailored, high-touch services that address multiple barriers and respond to individual patient needs. However, the optimal design, tailoring, and delivery methods for these interventions remain unclear due to the limited number of studies included. Thus, further research is needed, particularly theory-based RCTs with rigorous study designs that are conducted in countries beyond the USA and the UK.

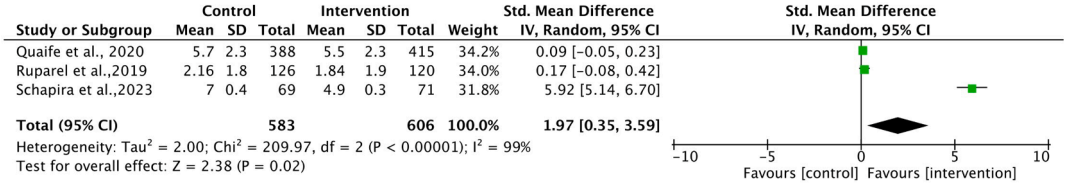


Fig. 6. Effects of intervention on lung cancer screening knowledge. CI, confidence interval.

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	4	placebo	Relative (95% CI)	Absolute (95% CI)		
Lung cancer screening uptake												
13	randomised trials	serious[a]	serious[b]	not serious	not serious	none	1162/4146 (28.0%)	1022/4821 (21.2%)	**OR 1.70** (1.10 to 2.62)	**102 more per 1,000** (from 16 more to 201 more)	⨁⨁◯◯ Low[ab]	
Lung cancer screening knowledge												
3	randomised trials	serious[c]	serious[b]	not serious	not serious	none	583	606	-	SMD **1.97 higher** (0.35 higher to 3.59 higher)	⨁⨁◯◯ Low[bc]	

CI: confidence interval; **OR:** odds ratio; **SMD:** standardised mean difference

Explanations

a. Downgrade one level for the risk of bias because of the high risk of bias in two studies.
b. Downgrade one level for the inconsistency because of the substantial heterogeneity
c. Downgrade one level for the risk of bias because of the high risk of bias in one study.

Fig. 7. Quality of evidence.

CRediT authorship contribution statement

Yuan Yang: Conceptualization, Methodology, Formal analysis, Visualization, Writing - Original Draft, Writing - Review & Editing. **Wenqian Zhao**: Methodology, Writing - Review & Editing. **Dorothy N. S. Chan**: Writing - Review & Editing. **Shishuang Zhou**: Methodology, Formal analysis. **Kai Chow Choi**: Methodology, Formal analysis. **Winnie K.W. So**: Conceptualization, Writing - Review & Editing, Supervision. All authors have read and approved the final manuscript.

Ethics statement

Not required.

Data availability statement

The data supporting the findings of this study are available within the article/supplementary material.

Declaration of generative AI and AI-assisted technologies in the writing process

No AI tools/services were used during the preparation of this work.

Funding

This study received no external funding.

Declaration of competing interest

The authors declare no conflict of interest. The corresponding author, Prof. Winnie K.W. So, is the Editor-in-Chief of *Asia–Pacific Journal of Oncology Nursing*. The other two authors, Prof. Dorothy N.S. Chan and Dr. Kai Chow Choi, serve as the editorial board members of *Asia–Pacific Journal of Oncology Nursing*. The article was subject to the journal's standard procedures, with peer review handled independently of Prof. Winnie K.W. So and their research groups.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apjon.2025.100746.

References

1. Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. May-Jun 2024;74(3):229–263. https://doi.org/10.3322/caac.21834.
2. Bi H, Tuo Y, Xiao X, et al. Observed and relative survival trends of lung cancer: a systematic review of population-based cancer registration data. *Thorac Cancer*. Nov 2023;15(2):142–151. https://doi.org/10.1111/1759-7714.15170.
3. Goldstraw P, Chansky K, Crowley J, et al. The IASLC lung cancer staging project: proposals for revision of the TNM stage groupings in the forthcoming (Eighth) edition of the TNM classification for lung cancer. *J Thorac Oncol*. Jan 2016;11(1):39–51. https://doi.org/10.1016/j.jtho.2015.09.009.
4. Leiter A, Veluswamy RR, Wisnivesky JP. The global burden of lung cancer: current status and future trends. *Nat Rev Clin Oncol*. Sep 2023;20(9):624–639. https://doi.org/10.1038/s41571-023-00798-3.
5. Becker N, Motsch E, Trotter A, et al. Lung cancer mortality reduction by LDCT screening-Results from the randomized German LUSI trial. *Int J Cancer*. Mar 2020;146(6):1503–1513. https://doi.org/10.1002/ijc.32486.
6. National Lung Screening Trial Research Team, Aberle DR, Adams AM, et al. Reduced lung cancer mortality with low-dose computed tomographic screening. *N Engl J Med*. 2011;365(5):395–409. https://doi.org/10.1056/NEJMoa1102873.
7. Bradley SH, Abraham S, Callister ME, et al. Sensitivity of chest X-ray for detecting lung cancer in people presenting with symptoms: a systematic review. *Br J Gen Pract*. Nov 2019;69(689):e827–e835. https://doi.org/10.3399/bjgp19X706853.
8. Canadian Task Force on Preventive Health Care. Recommendations on screening for lung cancer. *Can Med Assoc J*. 2016;188:425–432.
9. Japan Radiological Society. *The Japanese imaging guideline 2013*. Japan RadiologicalSociety; 2013. http://www.radiology.jp/content/files/diagnostic_imaging_guidelines_2013_e.pdf.
10. US Preventive Services Task Force, Krist AH, Davidson KW, et al. Screening for lung cancer: US preventive services task force recommendation statement. *JAMA*. Mar 2021;325(10):962–970. https://doi.org/10.1001/jama.2021.1117.
11. Fedewa SA, Bandi P, Smith RA, Silvestri GA, Jemal A. Lung cancer screening rates during the COVID-19 pandemic. *Chest*. 2022;161(2):586–589. https://doi.org/10.1016/j.chest.2021.07.030.
12. Cao W, Tan F, Liu K, et al. Uptake of lung cancer screening with low-dose computed tomography in China: a multi-centre population-based study. *eClinicalMedicine*. Jul 2022;52:101594. https://doi.org/10.1016/j.eclinm.2022.101594.
13. Ghimire B, Maroni R, Vulkan D, et al. Evaluation of a health service adopting proactive approach to reduce high risk of lung cancer: the Liverpool Healthy Lung Programme. *Lung Cancer*. 2019;134:66–71. https://doi.org/10.1016/j.lungcan.2019.05.026.
14. Crosbie PA, Balata H, Evison M, et al. Implementing lung cancer screening: baseline results from a community-based 'Lung Health Check' pilot in deprived areas of Manchester. *Thorax*. 2019;74(4):405–409. https://doi.org/10.1136/thoraxjnl-2017-211377.
15. *Cancer Control: Knowledge into Action: WHO Guide for Effective Programmes*. World Health Organization; 2007.
16. Cavers D, Nelson M, Rostron J, et al. Understanding patient barriers and facilitators to uptake of lung screening using low dose computed tomography: a mixed methods scoping review of the current literature. *Respir Res*. Dec 2022;23(1):374. https://doi.org/10.1186/s12931-022-02255-8.
17. Sedani AE, Davis OC, Clifton SC, Campbell JE, Chou AF. Facilitators and barriers to implementation of lung cancer screening: a framework-driven systematic review. *J Natl Cancer Inst*. Nov 2022;114(11):1449–1467. https://doi.org/10.1093/jnci/djac154.
18. Saab MM, FitzGerald S, Noonan B, et al. Promoting lung cancer awareness, help-seeking and early detection: a systematic review of interventions. *Health Promot Int*. Dec 2021;36(6):1656–1671. https://doi.org/10.1093/heapro/daab016.
19. Dodd RH, Sharman AR, McGregor D, et al. Education messages and strategies to inform the public, potential screening candidates and healthcare providers about lung cancer screening: a systematic review. *Prev Med*. Apr 2023;169:107459. https://doi.org/10.1016/j.ypmed.2023.107459.
20. Odole IP, Andersen M, Richman IB. Digital interventions to support lung cancer screening: a systematic review. *Am J Prev Med*. Jan 2024;66(5):899–908. https://doi.org/10.1016/j.amepre.2024.01.007.
21. Lin X, Lei F, Lin J, et al. Promoting lung cancer screen decision-making and early detection behaviors: a systematic review and meta-analysis. *Cancer Nurs*. 2024. https://doi.org/10.1097/NCC.0000000000001334. Advance online publication.
22. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. Mar 2021;372:n71. https://doi.org/10.1136/bmj.n71.
23. Akhtar N, Bansal JG. Risk factors of lung cancer in nonsmoker. *Curr Probl Cancer*. 2017;41(5):328–339. https://doi.org/10.1016/j.currproblcancer.2017.07.002.
24. McKenzie JE, Brennan SE, Ryan RE, Thomson HJ, Johnston RV, Thomas J. Chapter 3: defining the criteria for including studies and how they will be grouped for the synthesis [last updated August 2023]. In: Higgins JPT, Thomas J, Chandler J, et al., eds. *Cochrane Handbook for Systematic Reviews of Interventions*. Cochrane; 2024. version 6.5. www.training.cochrane.org/handbook.
25. Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Phys Ther*. 2009;89:873–880, 2009.
26. Fleiss JL. The statistical basis of meta-analysis. *Stat Methods Med Res*. 1993;2(2):121–145. https://doi.org/10.1177/096228029300200202.
27. Higgins JP, Thomas J, Chandler J, et al. *Cochrane Handbook for Systematic Reviews of Interventions*. John Wiley & Sons; 2019.
28. Higgins JPT. Green S. 10.4.3 Tests for funnel plot asymmetry. https://handbook-5-1.cochrane.org/chapter_10/10_4_3_tests_for_funnel_plot_asymmetry.htm; 2011. Accessed October 7, 2024.
29. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. Aug 2019;366:l4898. https://doi.org/10.1136/bmj.l4898.
30. McMaster University and Evidence Prime. GRADEpro guideline development tool (Computer software) https://www.gradepro.org/; 2021.
31. Adler DH, Wood N, Fiscella K, et al. Increasing uptake of lung cancer screening among emergency department patients: a pilot study. *J Emerg Med*. Aug 2024;67(2):e164–e176. https://doi.org/10.1016/j.jemermed.2024.03.034.
32. Baggett TP, Sporn N, Barbosa Teixeira J, et al. Patient navigation for lung cancer screening at a health care for the homeless program: a randomized clinical trial. *JAMA Intern Med*. Aug 2024;184(8):892–902. https://doi.org/10.1001/jamainternmed.2024.1662.
33. Begnaud AL, Joseph AM, Lindgren BR. Randomized electronic promotion of lung cancer screening: a pilot. *JCO Clin Cancer Inform*. Nov 2017;1:1–6. https://doi.org/10.1200/CCI.17.00033.
34. Bhalla S, Natchimuthu V, Lee JL, et al. Effect of patient navigation on completion of lung cancer screening in vulnerable populations. *J Natl Compr Cancer Netw*. Mar 2024;22(3):151–157. https://doi.org/10.6004/jnccn.2023.7094.
35. DiCarlo M, Myers P, Daskalakis C, et al. Outreach to primary care patients in lung cancer screening: a randomized controlled trial. *Prev Med*. Jun 2022;159:107069. https://doi.org/10.1016/j.ypmed.2022.107069.
36. Kathuria H, Gunawan A, Spring M, et al. Hospitalization as an opportunity to engage underserved individuals in shared decision-making for lung cancer screening: results

from two randomized pilot trials. *Cancer Causes Control*. Nov 2022;33(11): 1373–1380. https://doi.org/10.1007/s10552-022-01620-8.
37. Percac-Lima S, Ashburner JM, Rigotti NA, et al. Patient navigation for lung cancer screening among current smokers in community health centers a randomized controlled trial. *Cancer Med*. Mar 2018;7(3):894–902. https://doi.org/10.1002/cam4.1297.
38. Quaife SL, Ruparel M, Dickson JL, et al. Lung screen uptake trial (LSUT): randomized controlled clinical trial testing targeted invitation materials. *Am J Respir Crit Care Med*. Apr 2020;201(8):965–975. https://doi.org/10.1164/rccm.201905-0946OC.
39. Raz DJ, Ismail MH, Haupt EC, et al. Improving utilization of lung cancer screening through incorporating a video-based educational tool into smoking cessation counseling. *Clin Lung Cancer*. Mar 2021;22(2):83–91. https://doi.org/10.1016/j.cllc.2020.12.003.
40. Ruparel M, Quaife SL, Ghimire B, et al. Impact of a lung cancer screening information film on informed decision-making: a randomized trial. *Ann Am Thorac Soc*. Jun 2019;16(6):744–751. https://doi.org/10.1513/AnnalsATS.201811-841OC.
41. Schapira MM, Hubbard RA, Whittle J, et al. Lung cancer screening decision aid designed for a primary care setting: a randomized clinical trial. *JAMA Netw Open*. Aug 2023;6(8):e2330452. https://doi.org/10.1001/jamanetworkopen.2023.30452.
42. Volk RJ, Lowenstein LM, Leal VB, et al. Effect of a patient decision aid on lung cancer screening decision-making by persons who smoke: a randomized clinical trial. *JAMA Netw Open*. Jan 2020;3(1):e1920362. https://doi.org/10.1001/jamanetworkopen.2019.20362.
43. Rana T, Chan DNS, Nguyen KT, Choi KC, So WKW. Effectiveness of community health worker-led interventions in enhancing colorectal cancer screening pptake in racial and ethnic minority populations: a systematic review and meta-analysis. *Cancer Nurs*. Sep-Oct 2024;47(5):368–376. https://doi.org/10.1097/NCC.0000000000001222.
44. Nelson HD, Cantor A, Wagner J, et al. Effectiveness of patient navigation to increase cancer screening in populations adversely affected by health disparities: a meta-analysis. *J Gen Intern Med*. Oct 2020;35(10):3026–3035. https://doi.org/10.1007/s11606-020-06020-9.
45. Rodríguez-Gómez M, Ruiz-Pérez I, Martín-Calderón S, et al. Effectiveness of patient-targeted interventions to increase cancer screening participation in rural areas: a systematic review. *Int J Nurs Stud*. Jan 2020;101:103401. https://doi.org/10.1016/j.ijnurstu.2019.103401.
46. World Health Organization. *Guide to Cancer Early Diagnosis*. World Health Organization; 2017. https://iris.who.int/handle/10665/254500.
47. Fagan HB, Fournakis NA, Jurkovitz C, et al. Telephone-based shared decision-making for lung cancer screening in primary care. *J Cancer Educ*. Aug 2020;35(4): 766–773. https://doi.org/10.1007/s13187-019-01528-z.
48. Jallow M, Bonfield S, Kurtidu C, et al. Decision support tools for low-dose CT lung cancer screening: a scoping review of information content, format, and presentation methods. *Chest*. Oct 2020;162(4):930–941. https://doi.org/10.1016/j.chest.2021.12.638.